

- ① If $\mathbf{u}$ and $\mathbf{v}$ are two vectors in $\mathbb{R}^n$ with the same magnitude, then magnitude of $\mathbf{u} - \mathbf{v}$ must be zero. Is the above statement True or False? Justify.

$$1. \|v\| = \|u\|$$

Let's take an example:

$$\vec{v} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \rightarrow \|\vec{v}\| = 1$$

$$\vec{u} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \rightarrow \|\vec{u}\| = 1$$

$$\begin{aligned} \|\vec{v} - \vec{u}\| &= \sqrt{(v-u)^T (v-u)} \\ &= \sqrt{1+1} = \sqrt{2} \neq 0 \end{aligned}$$

$\therefore$ Statement is False

2. Can two orthogonal vectors be a part of the same space?

3. If $\mathbf{u} = \begin{bmatrix} -1 \\ -1 \\ -1 \end{bmatrix}$ is orthogonal to vectors $\mathbf{v}$ and $\mathbf{w}$, does that mean $\mathbf{v}$ and $\mathbf{w}$ are parallel?
Explain.

2. Yes.

Eg. $\vec{u} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ $\vec{v} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ $\vec{u}^T \vec{v} = 0$

$\therefore$ They are orthogonal and both belong to $\mathbb{R}^2$

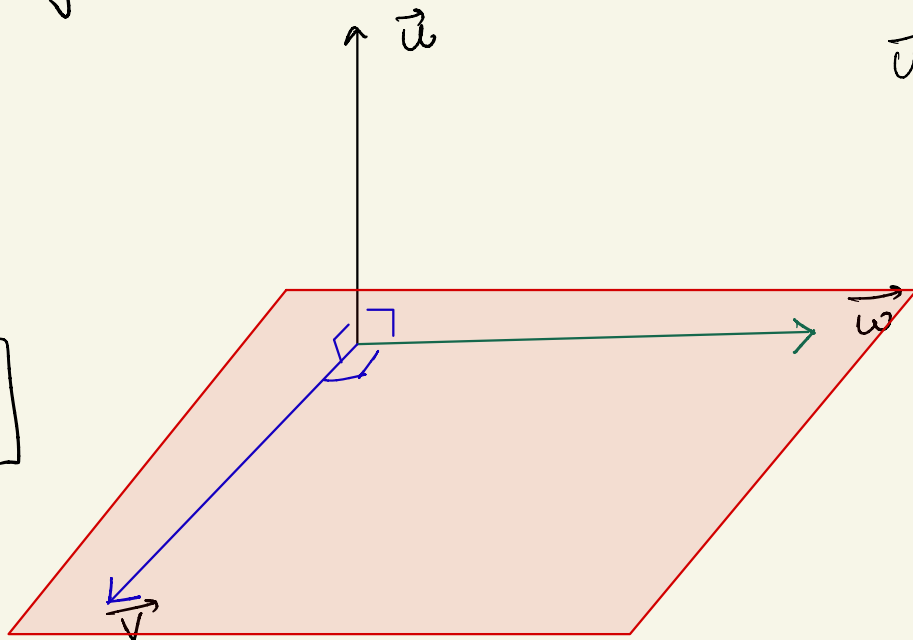
3. Given $\vec{u} \perp \vec{v}$
and $\vec{u} \perp \vec{w}$

Visually:

eg.

$$\vec{v} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

$$\vec{w} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$



$$\vec{u} \perp \vec{v}$$

$$\vec{u} \perp \vec{w}$$

but

$$\vec{v} \perp \vec{w}$$

∴ Statement is

False

* $\vec{u}$ is in $\mathbb{R}^3$. All vectors orthogonal to $\vec{u}$ are in a plane orthogonal to $\vec{u}$.
Two vectors can lie in the same plane and point in different directions.

4. Suppose $\mathbf{u}$ is orthogonal to vectors $\mathbf{v}$ and $\mathbf{w}$. Is $\mathbf{u}$ orthogonal to $2\mathbf{v} - 5\mathbf{w}$? Justify your answer.

4.

To check if $u^T (2v - 5w) = 0$

$$u^T (2v - 5w) = 2u^T v - 5u^T w$$

Since $u \perp v$, $u^T v = 0$

$u \perp w$, $u^T w = 0$

$\therefore 2u^T v = 0$, $5u^T w = 0$

$\therefore 2u^T v - 5u^T w = 0$

$\Rightarrow u^T (2v - 5w) = 0$

$\Rightarrow \vec{u} \perp (2\vec{v} - 5\vec{w})$

5. Suppose $\mathbf{u}$ is orthogonal to $(\mathbf{v} + \mathbf{w})$ and $(\mathbf{v} - \mathbf{w})$. Check if $\mathbf{u}$ is orthogonal to $\mathbf{v}$.

5.

Given $u \perp (v+w)$ and $u \perp (v-w)$
To check if $u \perp v$ i.e. if $u^T v = 0$

Given

$$\begin{aligned} u^T (v+w) &= 0 \rightarrow u^T v + u^T w = 0 \\ u^T (v-w) &= 0 \rightarrow u^T v - u^T w = 0 \end{aligned} \quad \left. \vphantom{\begin{aligned} u^T (v+w) &= 0 \\ u^T (v-w) &= 0 \end{aligned}} \right\} \text{adding them}$$

$$2 u^T v - 0 = 0$$
$$\rightarrow 2 u^T v = 0$$
$$\rightarrow u^T v = 0$$

$\therefore \vec{u}$ is orthogonal to $\vec{v}$

6. Suppose $\mathbf{u}$ is orthogonal to vectors $\mathbf{v}$ and $\mathbf{w}$. Prove that $\mathbf{u}$ is orthogonal to all linear combinations of $\mathbf{v}$ and $\mathbf{w}$.

6. Given $u \perp v$ and $u \perp w$
 $\Rightarrow u^T v = 0$ and $u^T w = 0$

To check if $u^T (c_1 v + c_2 w) = 0$

where c_1 and c_2 are scalar quantities

$$u^T (c_1 v + c_2 w)$$

$$= c_1 \underbrace{(u^T v)}_{\substack{\downarrow \\ 0}} + c_2 \underbrace{(u^T w)}_{\substack{\downarrow \\ 0}}$$

$$= 0 + 0 = 0$$

$\therefore u^T (c_1 v + c_2 w) = 0 \rightarrow u$ is orthogonal to
 all linear combinations of v and w .

7. Suppose $\mathbf{u}$ is orthogonal to vector $\mathbf{v}$ and $\mathbf{u} \cdot \mathbf{w} = 3$. For what value of c is $\mathbf{u}$ orthogonal to $(\mathbf{v} + c\mathbf{w})$?

7. given $u^T v = 0$
and $u^T w = 3$

To find scalar c such that $u^T (v + cw) = 0$

$$u^T (v + cw) = 0$$
$$\Rightarrow \underbrace{u^T v}_0 + c u^T w = 0$$

$$\Rightarrow c \underbrace{u^T w}_3 = 0$$

$$\Rightarrow 3c = 0$$

$$\Rightarrow c = 0$$

8. Suppose $\mathbf{u}$ is orthogonal to vector $\mathbf{v}$ and $\mathbf{w} = 3\mathbf{v} - 2\mathbf{u}$. Is $\mathbf{w}$ necessarily orthogonal to $\mathbf{u}$? Justify your answer.

8.

Given $u^T v = 0$

$$w = 3v - 2u$$

To check if $u^T w = 0$

$$\begin{aligned} u^T w &= u^T (3v - 2u) \\ &= 3 \underbrace{u^T v}_0 - 2 \underbrace{u^T u}_0 \end{aligned}$$

0 only if $\vec{u}$ is a zero vector

$\Rightarrow u^T w$ will be zero only if $\vec{u}$ is a zero vector. otherwise $u^T w \neq 0 \therefore$ For non-zero $\vec{u}$, $\vec{u}$ is not orthogonal to $\vec{w}$.